

ĐẠI HỌC
BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

INTRODUCTION TO DEEP LEARNING – PROJECT

SENTIMENT ANALYSIS OF AMAZON FINE FOOD REVIEWS

BiLSTM with attention versus fine-tuned BERT

Nguyễn Tuệ Lâm – 20261207M
Instructor: Dr. Trần Việt Trung
Hanoi University of Science and Technology
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The problem

Three-class sentiment from the text of a food review

Shops receive thousands of reviews a day; a seller needs the **sentiment** of each text to see which products disappoint.

- **Input:** title + body of an English review
- **Output:** negative, neutral or positive
- **Labels from stars** (weak labels, no human annotation):

Stars	1	2	3	4	5
Label	negative		neutral	positive	

Counting words is not enough

"Not as Advertised. Product arrived labeled as Jumbo Salted Peanuts...the peanuts were actually small sized unsalted." A negative review with no strongly negative word.

Why it is hard

- **Skewed labels:** 78% positive, 7.6% neutral
- **Ambiguous neutral:** 3-star reviews praise and complain at once
- **Weak labels:** 5 stars with a complaint exist
- **Context:** negation ("not bad"), contrast ("good but..."), sarcasm
- **Dirty data:** repeated reviews, HTML, very uneven length

Main metric: Macro-F1

Macro-F1 $= \frac{1}{3}\sum_c F1_c$, every class counts equally. "Always positive" scores **78% accuracy but only 0.29 Macro-F1**.

Data quality and cleaning

Amazon Fine Food Reviews: 568,454 raw rows, 10 columns, October 1999 – October 2012

Issues measured on the raw file

Issue	Rows	Share
Repeated review (key)	174,521	30.70%
Repeated `Text`	174,875	30.76%
Missing `Summary`	27	0.005%
Missing `ProfileName`	26	0.005%
Numerator > denominator	2	<0.001%

key = (`UserId`, `ProfileName`, `Time`, `Text`)

Why repeats matter

No row is duplicated in *all* columns: Amazon shares one review among product variants. User `A395BORC6FGVXV`'s 2-star "Cough Medicine" review appears under 3 product codes. A random split would put copies in both train and test: **data leakage**.

Cleaning pipeline (always keep the first occurrence)

Raw file: 568,454 rows

↓ −174,521

1. Drop repeated (user, name, time, text) → **393,933**

↓ −2

2. Drop numerator > denominator → **393,931**

↓ −354

3. Drop texts still identical to an earlier row → **393,577**

↓

4. Missing `Summary` → empty string; label from `Score`

Result: **393,577 reviews** (69.2% of raw). Step 3 also drops identical short reviews by different people: a deliberate trade of 0.09% of the data to close the last leakage route.

Model input and data split

Leakage-safe, one frozen configuration

Model input text

$\text{text} = \text{clean}(\texttt{Summary}) + \text{“. ”} + \text{clean}(\texttt{Text})$
clean = unescape HTML entities, delete tags and URLs, collapse spaces. **Not** lower-cased; punctuation, stop words and negations are kept (*not*, *no*, *never* decide sentiment).

Before: …`were bummed.<br /><br />Now, because…`
After: …`were bummed. Now, because…`

HTML in 26.1% of reviews, URLs in 2.1%, encoded characters in 0.3%. Cleaning all 393,577 reviews takes 19.5 s.

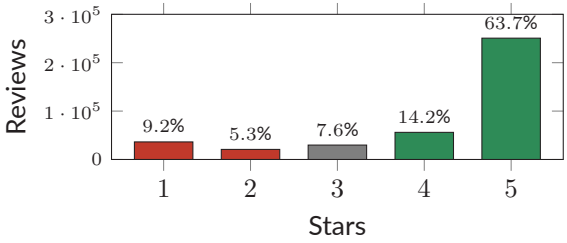
	Reviews	neg	neu	pos
Train (80%)	314,861	45,654	23,803	245,404
Validation (10%)	39,358	same shares		
Test (10%)	39,358	same shares		

- Stratified by label, seed 42 (14.5 / 7.6 / 77.9% in every part)
- Train updates weights; validation picks epoch and threshold; **test is read once**
- Vocabulary and class weights come from **train only**
- Class-weighted cross-entropy, $w_c = N/(K\,N_c)$:

	neg	neu	pos
w_c	2.299	4.409	0.428

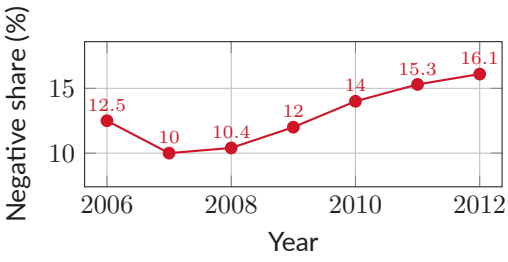
EDA 1: class balance and change over time

393,577 cleaned reviews



negative 14.5% neutral 7.6% positive 77.9%

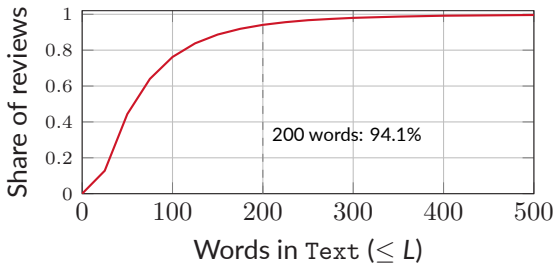
- Positive outnumbers neutral **10.3 : 1**; 5 stars alone are 63.7%
- $\Rightarrow$ stratified split, class weights, Macro-F1



- Volume grows exponentially from 2006 (15,752 in 2007, 139,085 in 2012); mean score falls 4.39 $\rightarrow$ 4.11
- The label distribution drifts; our split is random, so scores do *not* measure future reviews

EDA 2: text length

Right-skewed: median 57 words, longest review 3,432 words



	Mean	Median	P90	P99	Max
Text (words)	79.8	57	159	377	3,432
Summary (words)	4.1	3	8	12	42

Median words	1 star	2	3	4	5
	61	67	**70**	65	52

- Mixed-opinion reviews (2–3 stars) are the longest
- Titles are short but dense (“Great taffy”, “Not as Advertised”) ⇒ put the title first

Decisions

BiLSTM keeps **200 words** (6.47% of training reviews cut); BERT keeps **256 WordPiece tokens** (6.2% cut), keeping the beginning with the title. **Dynamic padding and length bucketing** so short reviews are not padded to a 3,000-word outlier.

EDA 3: vocabulary, noise, users and design decisions

Class-balanced sample of 20,000 reviews per class, stop words removed except negations

Top bigrams per class

negative	*not good*, *would not*, *not buy*, *waste money*
neutral	*not bad*, *not sure*, *not great*, *pretty good*
positive	*highly recommend*, *great product*, *really good*

Single words (*like*, *taste*, *good*, *not*) occur in *all* three classes; neutral has no vocabulary of its own.

Other findings

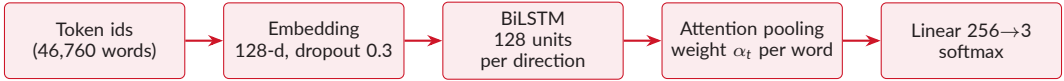
- 80.3% of users wrote one review; 78.7% of products have $\leq$5 reviews $\Rightarrow$ the model must read the text
- 1-star reviews get votes more often (74.7% vs 49.8% for 5 stars) but a lower helpful ratio (0.54 vs 0.90); unavailable when a review is posted $\Rightarrow$ **not a feature**

From findings to decisions

Finding	Decision
31% repeated reviews	dedupe before the split
78 / 14.5 / 7.6% labels	stratify, class weights, Macro-F1
Right-skewed length	200 words / 256 tokens, bucketing
Short, dense titles	input = title + body
26% HTML	light cleaning, keep negations
Context matters, pairs beat words	sequence model (BiLSTM) and Transformer (BERT)
Label drift over time	report as a limitation

Two models, one question: how much does pre-training help?

Same data, same class-weighted loss, same test reviews



BiLSTM + attention: from scratch

- 6.25 M parameters; reads 200 words in both directions
- AdamW (lr 2×10^{-3}), clipping 1.0, early stopping; best epoch 4 (val Macro-F1 0.7504); 3.4 min on a T4
- Length bucketing: 1,230 batches per epoch; padding positions masked in attention
- Attention weights make the model inspectable: they usually fall on *great*, *not*, *but*, *mistake*

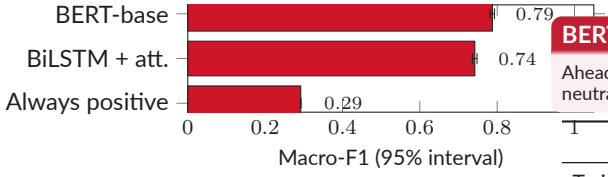
BERT-base: pre-trained, fine-tuned

- `bert-base-uncased`: 12 layers, 109.48 M parameters, pre-trained on Wikipedia + BookCorpus; 3-class head on `[CLS]`
- 256 WordPiece tokens, batch 32, 2 epochs, AdamW (lr 3×10^{-5}), 10% warm-up, fp16
- Val Macro-F1 0.7664 → **0.7864**, still rising: not saturated
- 64.4 min on a T4; one run, no hyperparameter search

Main results

39,358 test reviews; paired bootstrap, 1,000 resamples

Model	Accuracy	Macro-F1	F1 negative	F1 neutral	F1 positive
BERT-base	**0.9074**	**0.7877**	**0.8470**	**0.5540**	**0.9620**
BiLSTM + attention	0.8788	0.7422	0.7863	0.4916	0.9488
Always positive	0.7794	0.2920	0	0	0.8760



BERT − BiLSTM = +0.0454 [+0.0400, +0.0507]

Ahead in 100% of resamples and on every class (negative $+0.061$, neutral $+0.062$, positive $+0.013$).

	BERT-base	BiLSTM
Training / inference	64.4 min / 84 s	3.4 min / 4 s
Parameters	109.48 M	6.25 M

19$\times$ the training time for +0.0455 Macro-F1; the BiLSTM reaches 94% of BERT's score with 5% of its training time.

Where the models struggle

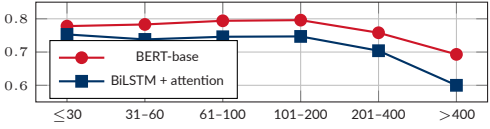
Neutral class, long reviews, noisy labels and sarcasm

Neutral: high recall, low precision

	True neutral found		Called neutral, is	
	neutral	other	neutral	other
BERT-base	68.1%	31.9%	46.7%	53.3%
BiLSTM	67.6%	32.4%	38.6%	61.4%

Neutral is over-predicted (the data and the 4.4× loss weight). A validation-chosen bias helps the BiLSTM (+0.013) but not BERT.

Macro-F1 by review length: lowest above 400 words (BiLSTM 0.600, BERT 0.693), where the end is cut



Confident errors of BERT

165 five-star reviews predicted *negative*, 58 one-star predicted *positive*, 948 neutral misclassified.

"Now I would not recommend it to my worst enemy. The coffee now has no aroma…" (5 stars): the text is right, the **label is noisy**. Not all are noise: *"My dogs won't eat anything else"* is praise that BERT misreads.

Sarcasm fools both

"Great, another box arrived completely crushed. Just what I needed." → positive (BiLSTM 87%, BERT 99%). Mixed sentences ("It's okay…") are labelled neutral by both.

Conclusions, limitations and future work

Conclusions

- **BERT-base is the better model:** Macro-F1 0.7877 vs 0.7422 (+0.0454, significant), better on every class
- Transfer learning beats learning from scratch here, at 19× the training time and 17.5× the parameters
- Neutral stays the hardest class (F1 0.49 and 0.55)

Limitations

- Weak labels: stars do not always match the text
- Random split: no test on future reviews
- Each model run once; BERT only 2 epochs (not saturated)
- One domain: Amazon food reviews, 1999–2012

Future work

- More epochs and seeds; RoBERTa / DeBERTa
- Time-based split; GloVe for the BiLSTM
- Human-labelled sample to measure label noise

THANK YOU

Questions & discussion